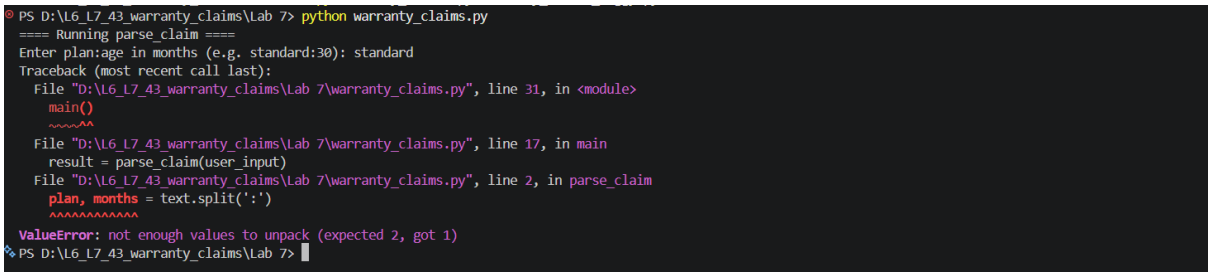
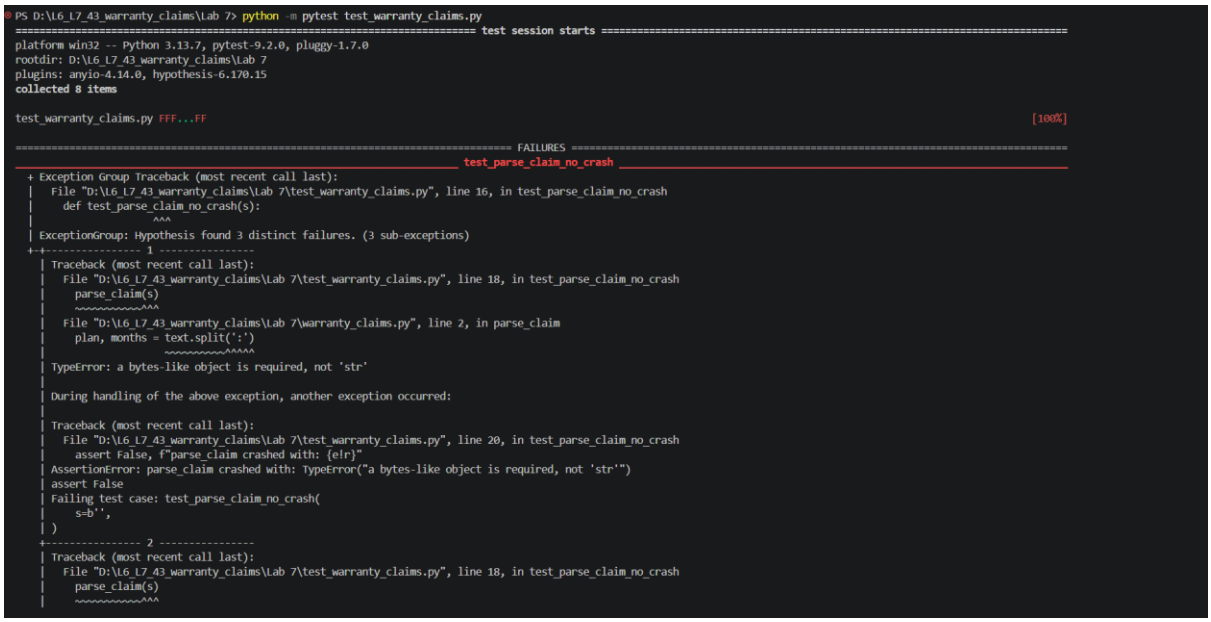
Screenshot 1: the buggy run



```
PS D:\L6_L7_43_warranty_claims\Lab 7> python warranty_claims.py
==== Running parse_claim ====
Enter plan:age in months (e.g. standard:30): standard
Traceback (most recent call last):
  File "D:\L6_L7_43_warranty_claims\Lab 7\warranty_claims.py", line 31, in <module>
    main()
    ~~~~^^
  File "D:\L6_L7_43_warranty_claims\Lab 7\warranty_claims.py", line 17, in main
    result = parse_claim(user_input)
  File "D:\L6_L7_43_warranty_claims\Lab 7\warranty_claims.py", line 2, in parse_claim
    plan, months = text.split(':')
    ^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)
PS D:\L6_L7_43_warranty_claims\Lab 7>
```

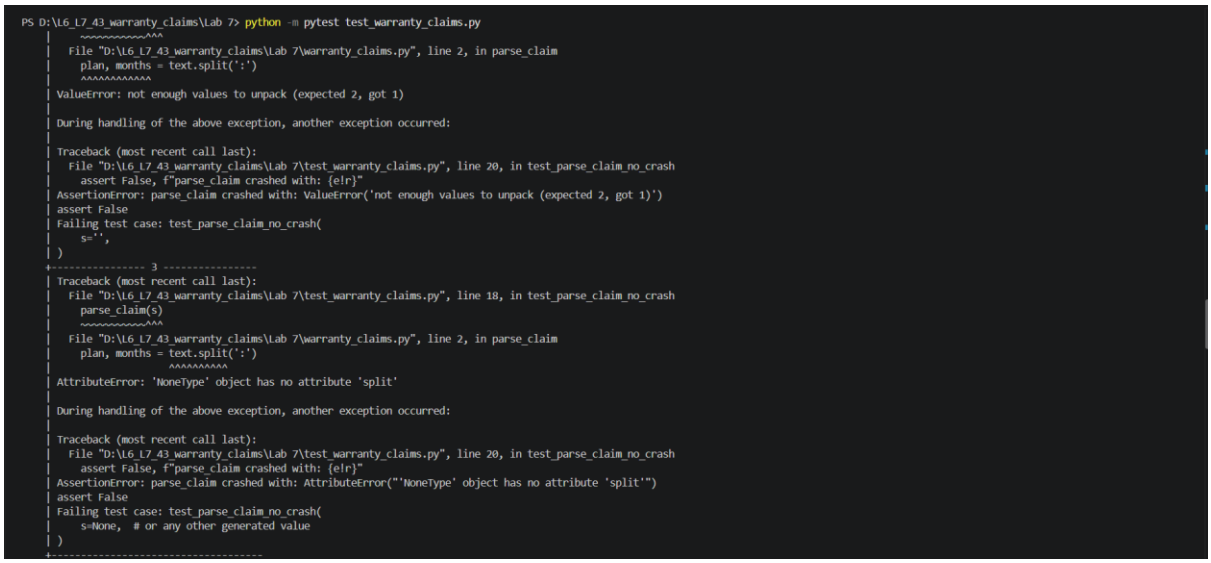
Screenshot 2: failing tests



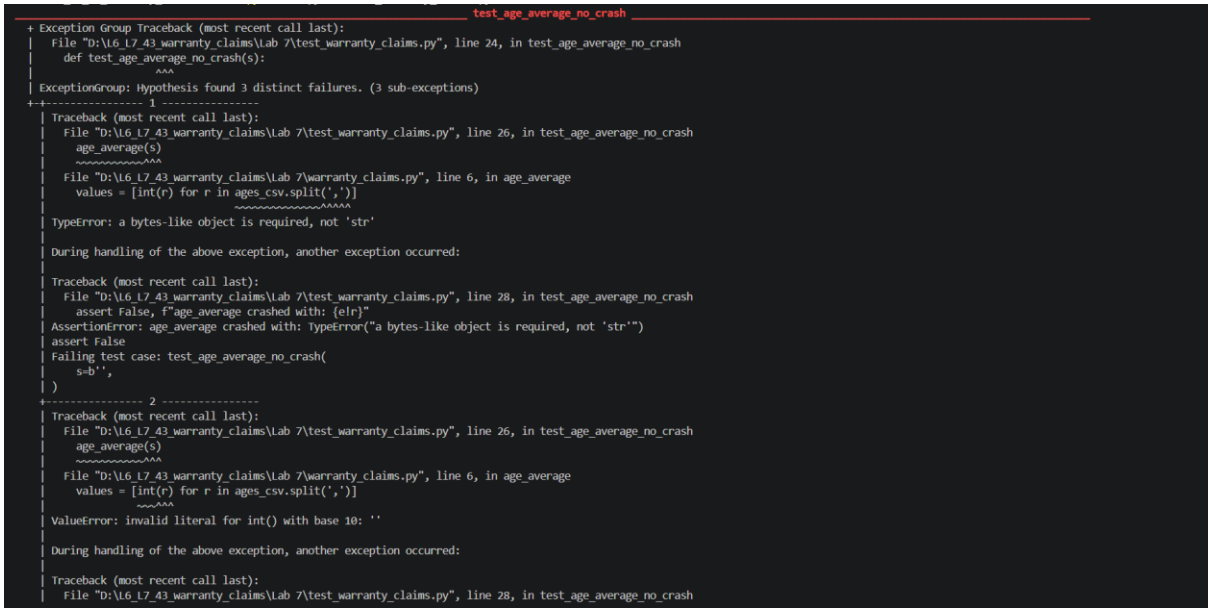
```
PS D:\L6_L7_43_warranty_claims\Lab 7> python -m pytest test_warranty_claims.py
============================= test session starts =============================
platform win32 -- Python 3.13.7, pytest-9.2.0, pluggy-1.7.0
rootdir: D:\L6_L7_43_warranty_claims\Lab 7
plugins: anyio-4.14.0, hypothesis-6.170.15
collected 8 items

test_warranty_claims.py FFF...FF                                         [100%]

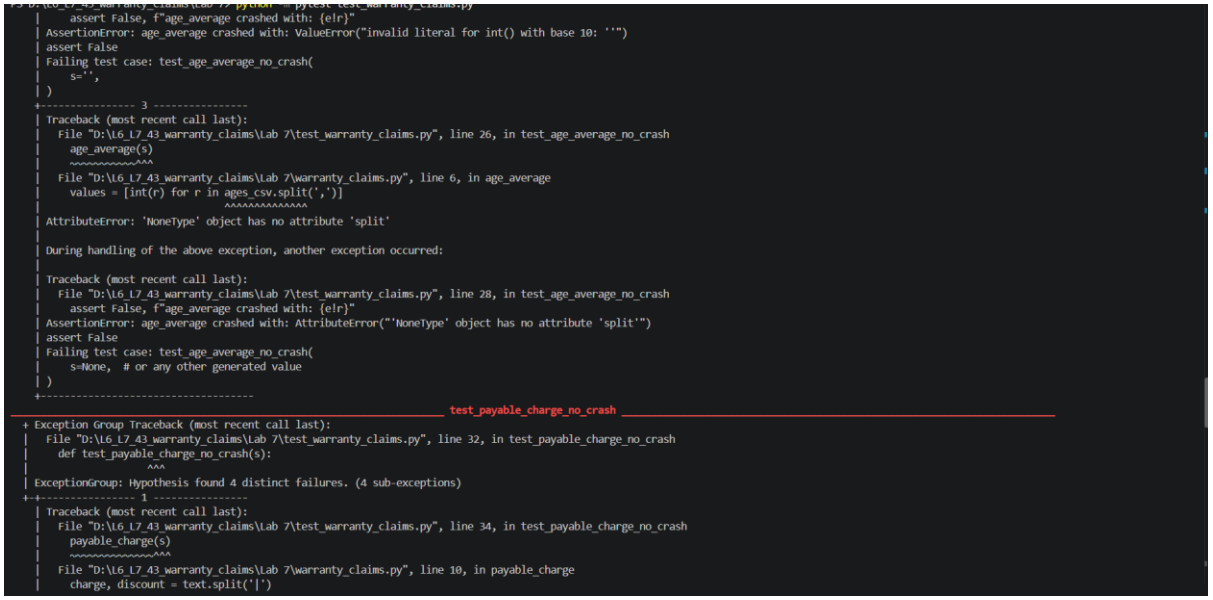
================================== FAILURES ===================================
_____________________ test_parse_claim_no_crash _____________________
  + Exception Group Traceback (most recent call last):
  |   File "D:\L6_L7_43_warranty_claims\Lab 7\test_warranty_claims.py", line 16, in test_parse_claim_no_crash
  |     def test_parse_claim_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\test_warranty_claims.py", line 18, in test_parse_claim_no_crash
    |     parse_claim(s)
    |     ~~~~~~~~~~~^^^
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\warranty_claims.py", line 2, in parse_claim
    |     plan, months = text.split(':')
    |                    ~~~~~~~~~~^^^^^
    | TypeError: a bytes-like object is required, not 'str'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\test_warranty_claims.py", line 20, in test_parse_claim_no_crash
    |     assert False, f"parse_claim crashed with: {e!r}"
    | AssertionError: parse_claim crashed with: TypeError("a bytes-like object is required, not 'str'")
    | assert False
    | Failing test case: test_parse_claim_no_crash(
    |     s=b'',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\test_warranty_claims.py", line 18, in test_parse_claim_no_crash
    |     parse_claim(s)
    |     ~~~~~~~~~~~^^^
```



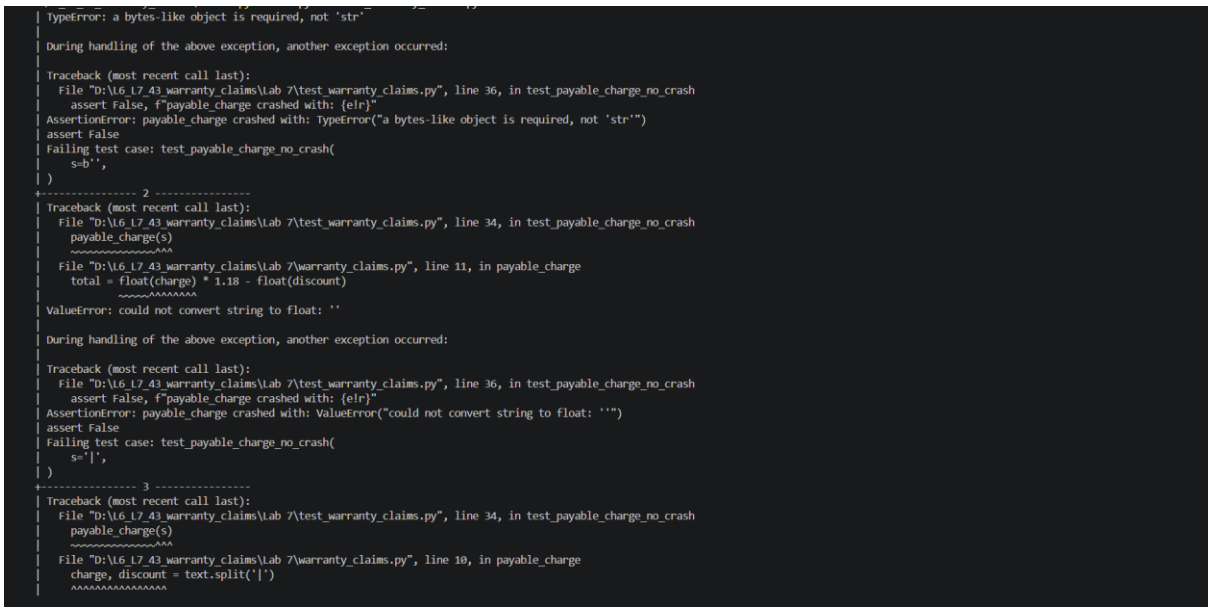
```
PS D:\L6_L7_43_warranty_claims\Lab 7> python -m pytest test_warranty_claims.py
    |     ~~~~~~~~~~~^^^
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\warranty_claims.py", line 2, in parse_claim
    |     plan, months = text.split(':')
    |     ^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\test_warranty_claims.py", line 20, in test_parse_claim_no_crash
    |     assert False, f"parse_claim crashed with: {e!r}"
    | AssertionError: parse_claim crashed with: ValueError('not enough values to unpack (expected 2, got 1)')
    | assert False
    | Failing test case: test_parse_claim_no_crash(
    |     s='',
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\test_warranty_claims.py", line 18, in test_parse_claim_no_crash
    |     parse_claim(s)
    |     ~~~~~~~~~~~^^^
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\warranty_claims.py", line 2, in parse_claim
    |     plan, months = text.split(':')
    |                    ^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\test_warranty_claims.py", line 20, in test_parse_claim_no_crash
    |     assert False, f"parse_claim crashed with: {e!r}"
    | AssertionError: parse_claim crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_parse_claim_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
```



```
_____ test_age_average_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "D:\L6_L7_43_warranty_claims\Lab 7\test_warranty_claims.py", line 24, in test_age_average_no_crash
  |     def test_age_average_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\test_warranty_claims.py", line 26, in test_age_average_no_crash
    |     age_average(s)
    |     ^^^^^^^^^^^^^^
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\warranty_claims.py", line 6, in age_average
    |     values = [int(r) for r in ages_csv.split(',')]
    |                              ^^^^^^^^^^^^^^^^^^^^
    | TypeError: a bytes-like object is required, not 'str'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\test_warranty_claims.py", line 28, in test_age_average_no_crash
    |     assert False, f"age_average crashed with: {e!r}"
    | AssertionError: age_average crashed with: TypeError("a bytes-like object is required, not 'str'")
    | assert False
    | Failing test case: test_age_average_no_crash(
    |     s=b'',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\test_warranty_claims.py", line 26, in test_age_average_no_crash
    |     age_average(s)
    |     ^^^^^^^^^^^^^^
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\warranty_claims.py", line 6, in age_average
    |     values = [int(r) for r in ages_csv.split(',')]
    |                 ^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\test_warranty_claims.py", line 28, in test_age_average_no_crash
```



```
    |     assert False, f"age_average crashed with: {e!r}"
    | AssertionError: age_average crashed with: ValueError("invalid literal for int() with base 10: ''")
    | assert False
    | Failing test case: test_age_average_no_crash(
    |     s='',
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\test_warranty_claims.py", line 26, in test_age_average_no_crash
    |     age_average(s)
    |     ^^^^^^^^^^^^^^
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\warranty_claims.py", line 6, in age_average
    |     values = [int(r) for r in ages_csv.split(',')]
    |                               ^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\test_warranty_claims.py", line 28, in test_age_average_no_crash
    |     assert False, f"age_average crashed with: {e!r}"
    | AssertionError: age_average crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_age_average_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____ test_payable_charge_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "D:\L6_L7_43_warranty_claims\Lab 7\test_warranty_claims.py", line 32, in test_payable_charge_no_crash
  |     def test_payable_charge_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 4 distinct failures. (4 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\test_warranty_claims.py", line 34, in test_payable_charge_no_crash
    |     payable_charge(s)
    |     ^^^^^^^^^^^^^^^^^
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\warranty_claims.py", line 10, in payable_charge
    |     charge, discount = text.split('|')
```



```
    | TypeError: a bytes-like object is required, not 'str'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\test_warranty_claims.py", line 36, in test_payable_charge_no_crash
    |     assert False, f"payable_charge crashed with: {e!r}"
    | AssertionError: payable_charge crashed with: TypeError("a bytes-like object is required, not 'str'")
    | assert False
    | Failing test case: test_payable_charge_no_crash(
    |     s=b'',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\test_warranty_claims.py", line 34, in test_payable_charge_no_crash
    |     payable_charge(s)
    |     ^^^^^^^^^^^^^^^^^
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\warranty_claims.py", line 11, in payable_charge
    |     total = float(charge) * 1.18 - float(discount)
    |             ^^^^^^^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\test_warranty_claims.py", line 36, in test_payable_charge_no_crash
    |     assert False, f"payable_charge crashed with: {e!r}"
    | AssertionError: payable_charge crashed with: ValueError("could not convert string to float: ''")
    | assert False
    | Failing test case: test_payable_charge_no_crash(
    |     s='|',
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\test_warranty_claims.py", line 34, in test_payable_charge_no_crash
    |     payable_charge(s)
    |     ^^^^^^^^^^^^^^^^^
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\warranty_claims.py", line 10, in payable_charge
    |     charge, discount = text.split('|')
    |     ^^^^^^^^^^^^^^^^
```

```
    |     ^^^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\test_warranty_claims.py", line 36, in test_payable_charge_no_crash
    |     assert False, f"payable_charge crashed with: {e!r}"
    | AssertionError: payable_charge crashed with: ValueError('not enough values to unpack (expected 2, got 1)')
    | assert False
    | Failing test case: test_payable_charge_no_crash(
    |     s='',
    | )
    +---------------- 4 ----------------
    | Traceback (most recent call last):
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\test_warranty_claims.py", line 34, in test_payable_charge_no_crash
    |     payable_charge(s)
    |     ^^^^^^^^^^^^^^^^^
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\warranty_claims.py", line 10, in payable_charge
    |     charge, discount = text.split('|')
    |                        ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\L6_L7_43_warranty_claims\Lab 7\test_warranty_claims.py", line 36, in test_payable_charge_no_crash
    |     assert False, f"payable_charge crashed with: {e!r}"
    | AssertionError: payable_charge crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_payable_charge_no_crash(
    |     s=None,  # or any other generated value
    | )
    +-----------------------------------
_____________________________________________ test_known_bad_inputs_return_none _____________________________________________

    def test_known_bad_inputs_return_none():
>       assert parse_claim("standard") is None        # no colon
               ^^^^^^^^^^^^^^^^^^^^^^^^
```

```
test_warranty_claims.py:70:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

text = 'standard'

    def parse_claim(text):
>       plan, months = text.split(':')
        ^^^^^^^^^^^^
E       ValueError: not enough values to unpack (expected 2, got 1)

warranty_claims.py:2: ValueError
_____________________________________________ test_known_good_inputs _____________________________________________

    def test_known_good_inputs():
        assert parse_claim(" standard :30") == ("Standard", 30.0)
>       assert age_average("12,24,37") == 24.33      # no floor-division truncation
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
E       AssertionError: assert 24 == 24.33
E        +  where 24 = age_average('12,24,37')

test_warranty_claims.py:90: AssertionError
=============================================================================== warnings summary ================================================================================
C:\Users\Prajwal\AppData\Roaming\Python\Python313\site-packages\anyio\pytest_plugin.py:14
  C:\Users\Prajwal\AppData\Roaming\Python\Python313\site-packages\anyio\pytest_plugin.py:14: PytestRemovedIn10Warning: _pytest.python.CallSpec2 has been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
============================================================================= short test summary info ===============================================================================
FAILED test_warranty_claims.py::test_parse_claim_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_warranty_claims.py::test_age_average_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_warranty_claims.py::test_payable_charge_no_crash - ExceptionGroup: Hypothesis found 4 distinct failures. (4 sub-exceptions)
FAILED test_warranty_claims.py::test_known_bad_inputs_return_none - ValueError: not enough values to unpack (expected 2, got 1)
FAILED test_warranty_claims.py::test_known_good_inputs - AssertionError: assert 24 == 24.33
========================================================== 5 failed, 3 passed, 1 warning in 8.96s ===========================================================
PS D:\L6_L7_43_warranty_claims\Lab 7>
```

Screenshot 3: the fixed run

Valid input:-

```
PS D:\L6_L7_43_warranty_claims\Lab 7> python warranty_claims.py
==== Running parse_claim ====
Enter plan:age in months (e.g. standard:30): standard:30
Plan and Age: ('Standard', 30.0)

==== Running age_average ====
Enter product ages as CSV (e.g. 12,24,36): 12,24,36
Average Age: 24.0

==== Running payable_charge ====
Enter charge|discount (e.g. 1500|200): 1500|200
Payable Charge: 1570.0
PS D:\L6_L7_43_warranty_claims\Lab 7>
```

Invalid input:-

```
PS D:\L6_L7_43_warranty_claims\Lab 7> python warranty_claims.py
==== Running parse_claim ====
Enter plan:age in months (e.g. standard:30): standard
Plan and Age: Invalid input

==== Running age_average ====
Enter product ages as CSV (e.g. 12,24,36): 1,,3
Average Age: Invalid input

==== Running payable_charge ====
Enter charge|discount (e.g. 1500|200): 1500
Payable Charge: Invalid input
PS D:\L6_L7_43_warranty_claims\Lab 7>
```

Screenshot 4: passing tests

```
PS D:\L6_L7_43_warranty_claims\Lab 7> python warranty_claims.py
==== Running parse_claim ====
Enter plan:age in months (e.g. standard:30): standard
Plan and Age: Invalid input

==== Running age_average ====
Enter product ages as CSV (e.g. 12,24,36): 1,,3
Average Age: Invalid input

==== Running payable_charge ====
Enter charge|discount (e.g. 1500|200): 1500
Payable Charge: Invalid input
PS D:\L6_L7_43_warranty_claims\Lab 7> python -m pytest test_warranty_claims.py
============================= test session starts =============================
platform win32 -- Python 3.13.7, pytest-9.2.0, pluggy-1.7.0
rootdir: D:\L6_L7_43_warranty_claims\Lab 7
plugins: anyio-4.14.0, hypothesis-6.170.15
collected 8 items

test_warranty_claims.py ........                                         [100%]

============================== warnings summary ===============================
C:\Users\Prajwal\AppData\Roaming\Python\Python313\site-packages\anyio\pytest_plugin.py:14
  C:\Users\Prajwal\AppData\Roaming\Python\Python313\site-packages\anyio\pytest_plugin.py:14: PytestRemovedIn10Warning: _pytest.python.CallSpec2 has been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
======================== 8 passed, 1 warning in 5.05s =========================
PS D:\L6_L7_43_warranty_claims\Lab 7>
```